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SECTION: G

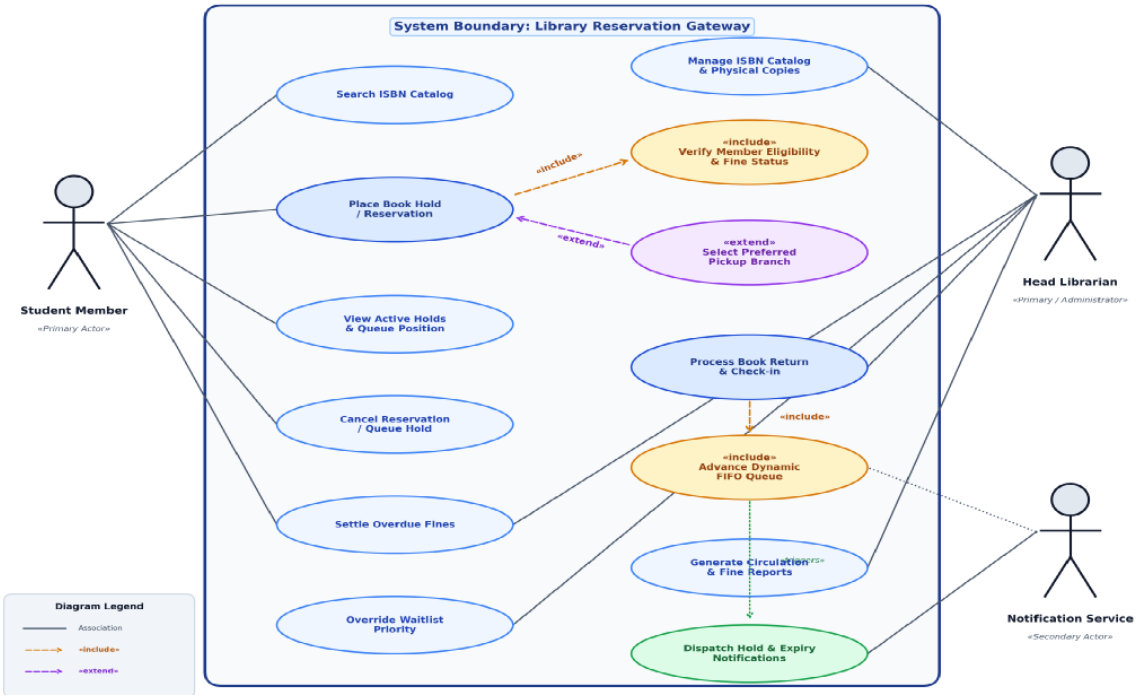
FUNCTIONAL REQUIREMENTS TABLE:

ID & Priority	Requirement Description	Acceptance Criteria	Rationale & Business Value
FR-001 High Type: Booking	Catalog Search & Waitlist Hold Placement: The system shall allow student members to search the ISBN catalog and place hold requests on currently loaned copies to join a dynamic FIFO reservation queue.	Pass: Eligible student is assigned next sequential integer position in FIFO queue; reservation token generated. Fail: Student holding >3 active holds or with fines >Rs 50 places hold.	Enforces fair, transparent, and strictly automated allocation of high-demand textbooks.
FR-002 High Type: Queue	Dynamic FIFO Queue Reallocation: The system shall automatically advance the FIFO waitlist and allocate an available copy to the #1 queued student upon book return check-in or hold expiry.	Pass: On return check-in, system flags book as 'Reserved-Ready', locks it for #1 student, and initiates 48h pickup window. Fail: Copy is allocated out-of-order or remains idle.	Minimizes shelf idle latency and eliminates manual librarian intervention in queue matching
FR-003 Medium Type: Alert	Multi-Channel Notification Dispatch: The system shall dispatch real-time email and SMS notifications when a held book is ready for collection, approaching expiration (24h warning), or overdue.	Pass: Notification dispatched within 60s of state change with pickup deadline and shelf location code. Fail: Dispatch failure occurs without retry log.	Maximizes pickup compliance and accelerates circulation turnover of library inventory.
FR-004 High Type: Finance	Overdue Fine Calculation & Settlement: The system shall automatically accrue daily overdue fines (Rs 10/day post-grace period), restrict borrowing/holding for delinquent accounts, and log settlement upon payment.	Pass: Fine accrues deterministically at 00:00 hrs; account automatically unblocks immediately upon full settlement. Fail: Delinquent student is permitted to place holds or checkout books.	Incentivizes timely resource returns and maintains an immutable institutional audit ledger.
FR-005 Medium Type: Admin	Catalog Inventory & Queue Override: The system shall allow the Head Librarian to add/update ISBN catalog records, inspect live waitlists, manually override queue priority with mandatory audit logging, and export analytics.	Pass: Librarian modifies inventory, reorders queue with logged administrative justification, and exports CSV/PDF. Fail: Priority override executed without recorded reason.	Provides operational flexibility for emergency academic needs while preserving audit traceability.

NON-FUNCTIONAL REQUIREMENTS:

ID & Type	Requirement Description	Acceptance Criteria	Rationale
NFR-001 Performance & Scalability Priority high	The catalog search query shall return paginated results for 50,000+ titles in.	Pass: latency 150 ms Fail: Latency >250 ms or dropped HTTP connections during peak load.	Prevents campus network saturation during peak semester rush and ensures a smooth, responsive student UX.
NFR-002 Security & Data Integrity Priority high	The system shall enforce Role-Based Access Control (RBAC),JWT token authentication	Pass: All API endpoints validate cryptographic signatures; zero unauthorized privilege escalation. Fail: unauthorized login	Guarantees reservation queue integrity against tampering, secures institutional financial records.

UML USE-CASE DIAGRAM:



USE-CASE FLOW SPECIFICATION:

Primary Actor	Student Member (Enrolled University Student)
Secondary Actors	Head Librarian (Administrative Monitor), Automated Notification Service (Message Dispatcher)
Brief Description	Allows an authenticated student member to place a reservation hold on a physical book whose copies are all currently on loan, entering a dynamic FIFO queue.
Preconditions	1. Student is authenticated via University SSO. 2. Target book exists in catalog and has 0 available shelf copies (all copies on loan). 3. Student has fewer than 3 active holds. 4. Student account has zero overdue blocks and outstanding fine balance <Rs 50.
Postconditions	Success Guarantee: Student is assigned sequential position in book's FIFO queue; status set to 'WAITING'; active hold count incremented; confirmation token generated. Minimal Guarantee: No queue state change occurs; user receives descriptive error message
Trigger	Student clicks the 'Place Hold / Reserve Copy' button on a borrowed book's catalog page
Main Success Scenario	1. Student searches the catalog and navigates to the detail page of a currently loaned title. 2. Student clicks 'Place Hold / Reserve Copy'. 3. System executes «include» Verify Member Eligibility & Fine Status to validate active hold quota and fine balance 4. System queries active loan records and calculates estimated availability date based on earliest due date. 5. System invokes «extend» Select Preferred Pickup Branch; Student designates target campus library desk. 6. Student reviews reservation summary and confirms submission. 7. System obtains atomic database lock on queue record, appends Student ID to FIFO waitlist, and assigns sequential queue position. 8. System increments student's active hold counter and generates a unique Reservation Reference ID. 9. System displays confirmation dialog containing Queue Position, Estimated Pickup Window, and Cancellation Option. 10. System asynchronously logs the transaction to the secure audit trail.
Alternate Flow: Hold Limit Reached	1. System detects student currently has 3 active reservations. 2. System displays error notification: 'Hold Quota Exceeded (Max 3 Active Holds Allowed). Please cancel an existing hold or return borrowed titles.' 3. System aborts reservation flow and returns student to catalog page without modifying queue state.